

Introduction to logistic regression

Agenda

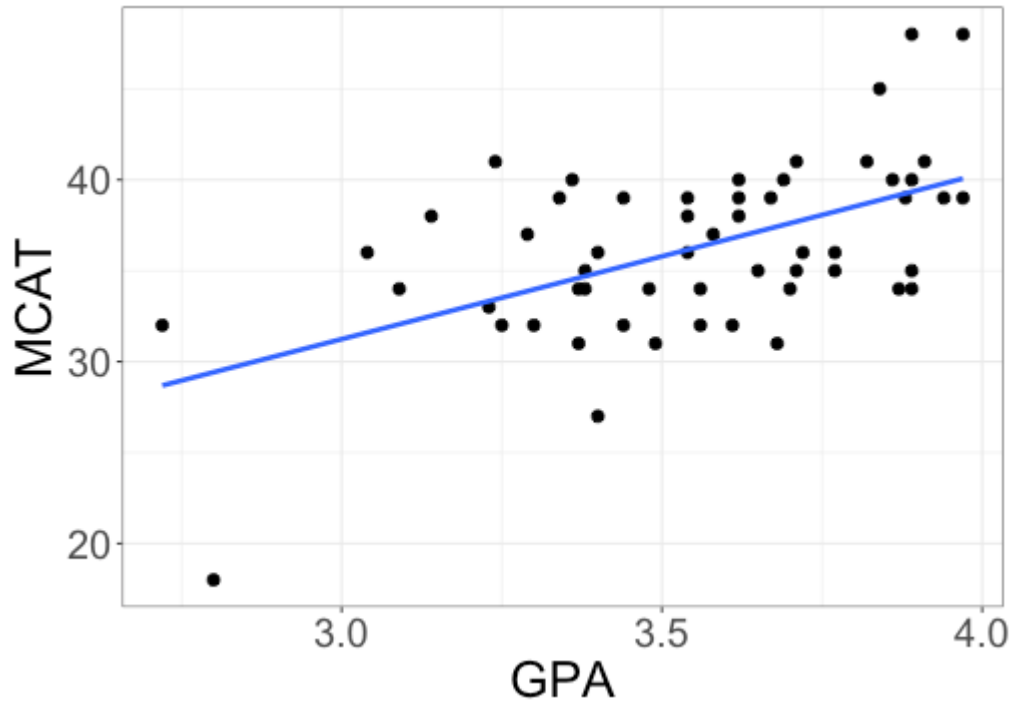
- + Project Part 3 due on Wednesday
- + Today: begin logistic regression

Data

55 students applying to medical school, with the following information on each student:

- + Acceptance: whether the student was accepted to medical school (1 = accepted, 0 = denied)
- + GPA: the student's college GPA
- + MCAT: the student's MCAT score

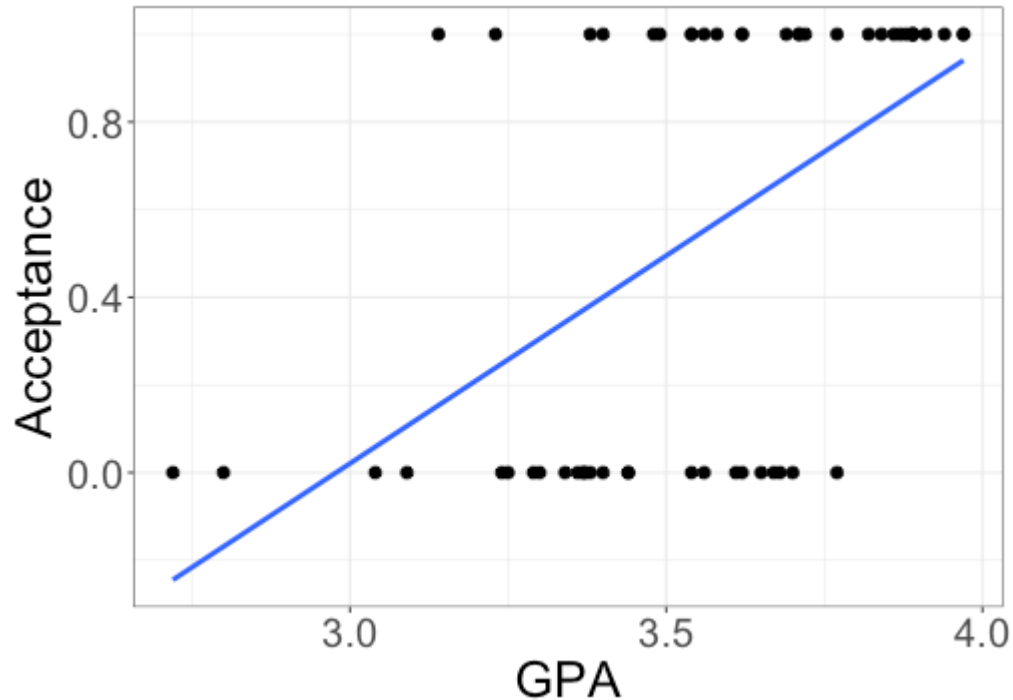
Linear regression



$$MCAT = \beta_0 + \beta_1 GPA + \varepsilon$$

What if our response isn't quantitative?

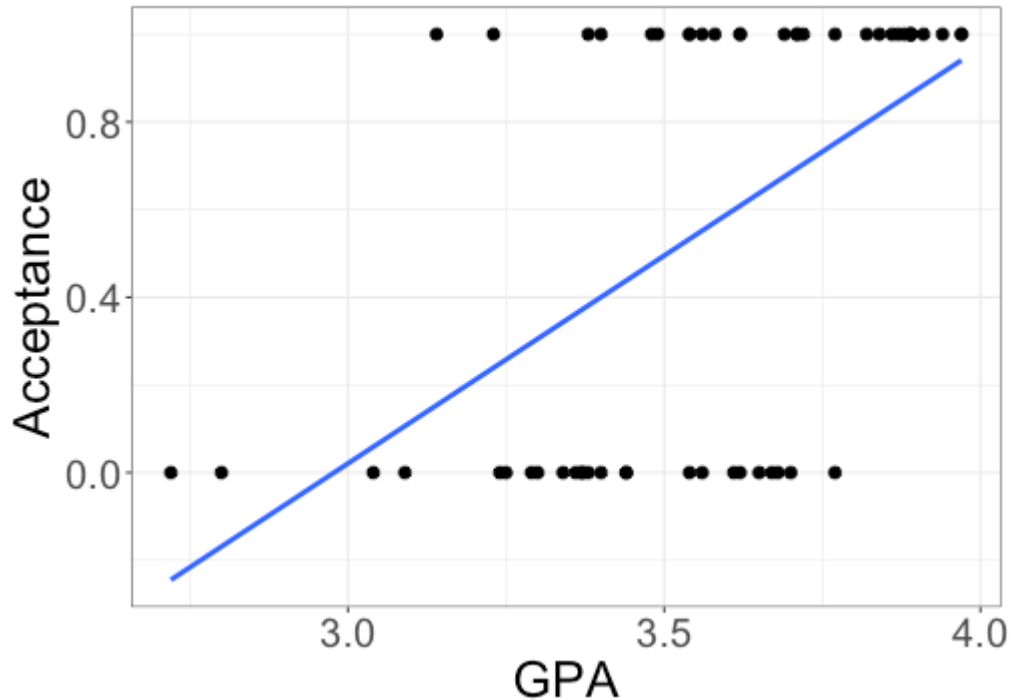
Binary response



$$\underbrace{\text{Acceptance}}_{\text{either 0 or 1}} \stackrel{?}{=} \underbrace{\beta_0 + \beta_1 \text{GPA} + \varepsilon}_{\text{not just 0 or 1 (whole range of different values)}}$$

What are the issues with this model?

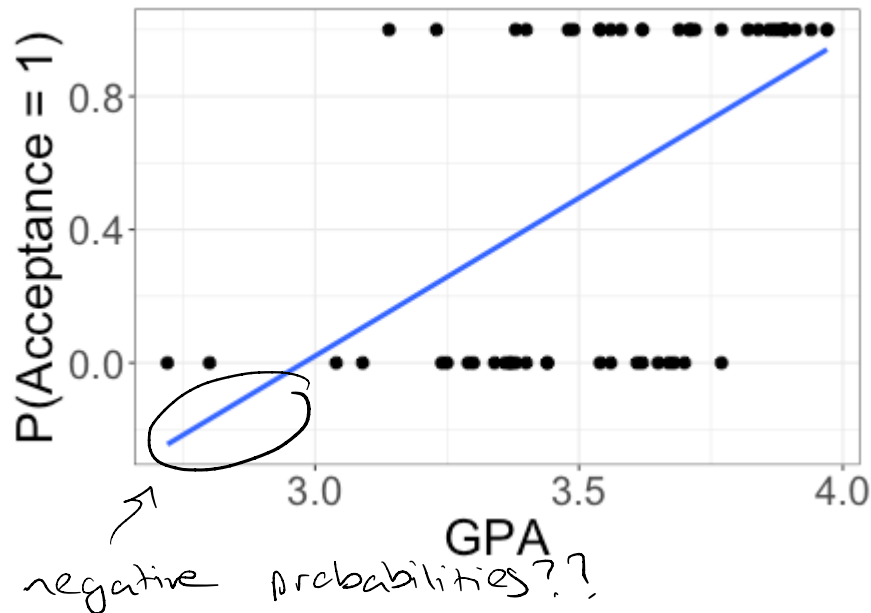
Issues with a binary response



- + Acceptance is either 0 or 1 -- we can't have any other values

How could we change the model to address this issue?

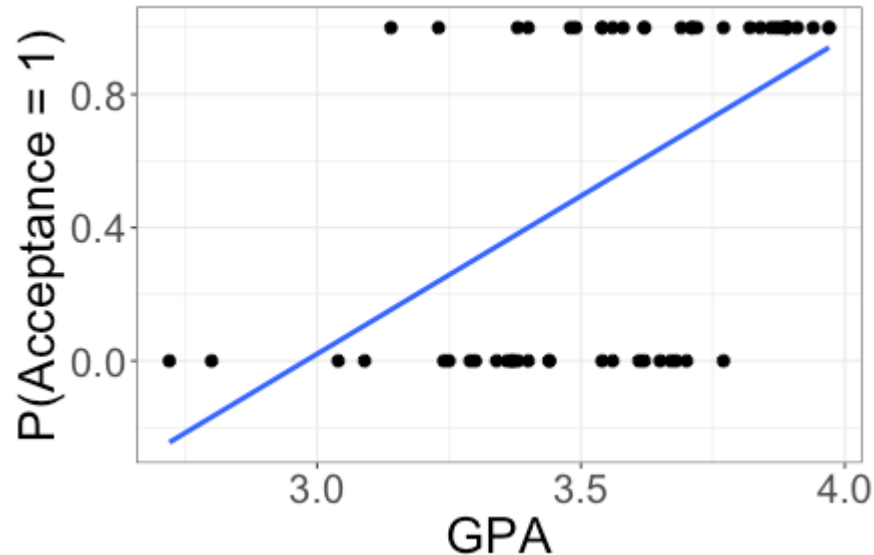
Working with probabilities



- + $P(\text{Acceptance} = 1)$ = probability student is accepted
- + $P(\text{Acceptance} = 1) \stackrel{?}{=} \beta_0 + \beta_1 \text{GPA}$

Does this fix our issues with the model?

Issues with probabilities



- + probabilities (e.g., $P(\textit{Acceptance} = 1)$) are between 0 and 1
- + Lines are never constrained, unless the slope is 0

Exploring binary outcomes

How can I summarize binary variables?

- + Probabilities
- + Odds
- + Odds ratios

Probabilities

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

What is the probability a TMS patient was pain free after two hours?

Probabilities

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

What is the probability a TMS patient was pain free after two hours?

$$39/100 = 0.39$$

Odds

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

Comparing probability of being pain free to probability of not being pain free:

+ $P(\text{TMS patient is pain free}) = 0.39$

+ $P(\text{TMS patient is not pain free}) = 0.61$

Odds that a TMS patient is pain free after two hours:

Probability of not being pain free is 1.56 x probability of being pain free (56% higher)

$$\frac{0.39}{0.61} = 0.64$$

probability of being pain free is 0.64 x probability of not being pain free 12/41

Odds

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

Odds that a TMS patient is pain free after two hours:

$$\frac{0.39}{0.61} = 0.64$$

- ✚ The probability of being pain free is 0.64 times the probability of not being pain free, for TMS patients

Odds

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

Odds that a TMS patient is pain free after two hours:

$$\frac{0.39}{0.61} = 0.64$$

+ Odds of being pain free = $\frac{\text{probability pain free}}{1 - \text{probability pain free}}$

Odds

Definition: Let π denote the probability an event happens (e.g., $\pi = P(\textit{Acceptance} = 1)$). Then the *odds* are

$$\frac{\pi}{1 - \pi}$$

✚ Note that here π is a probability, not the number 3.14159...

Odds

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

What are the odds a placebo patient was pain free after 2 hours?

$$\frac{22/100}{78/100} = \frac{0.22}{0.78}$$

$$= \frac{22}{78} = 0.28$$

Odds

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

What are the odds a placebo patient was pain free after 2 hours?

$$\frac{0.22}{1 - 0.22} = \frac{0.22}{0.78} = 0.28$$

Odds ratios

	TMS	Placebo	Total
Pain free two hours later	39	22	61
Not pain free two hours later	61	78	139
Total	100	100	200

Comparings odds:

- + Odds that a TMS patient is pain free after 2 hours: 0.64
- + Odds that a placebo patient is pain free after 2 hours: 0.28

Odds ratio: $\frac{0.64}{0.28} = 2.29$

- + The odds of being pain free after 2 hours are 2.29 times as high for TMS patients as for placebo patients

Class activity

Work on the class activity (handout).

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

What was the probability of survival for first class passengers?

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

What was the probability of survival for first class passengers?

$$136/216 = 0.63$$

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

What were the odds of survival for first-class passengers?

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

What were the odds of survival for first-class passengers?

$$(136/216)/(80/216) = 136/80 = 1.7$$

How would I interpret these odds?

probability of survival for 1st class passengers is 1.7 x probability of not surviving, i.e. for every 1st class passenger who did not survive, 1.7 survived

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

What were the odds of survival for first-class passengers?

$$(136/216)/(80/216) = 136/80 = 1.7$$

How would I interpret these odds?

For first class passengers, the probability of survival was 1.7 times the probability of not surviving.

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

Calculate the odds ratio comparing odds of survival for first-class passengers to odds of survival for second and third class passengers.

odds for 2nd/3rd class:

$$\frac{206/675}{469/675} = \frac{206}{469} = 0.44$$

$$\text{odds ratio} = \frac{1.7}{0.44} \approx 3.86$$

Class activity

	First class	Second and third class	Total
Survived	136	206	342
Did not survive	80	469	549
Total	216	675	891

Calculate the odds ratio comparing odds of survival for first-class passengers to odds of survival for second and third class passengers.

$$1.7/0.44 = 3.86$$

So the odds of survival are 3.86 times higher for first-class passengers than for second and third class passengers

Odds and probabilities

Definition: Let π denote the probability an event happens (e.g., $\pi = P(\textit{Acceptance} = 1)$). Then the *odds* are

$$\frac{\pi}{1 - \pi}$$

✚ Note that here π is a probability, not the number 3.14159...

If the probability of an event is > 0.5 , what is true about the odds?

(A) Odds > 0.5

(B) Odds > 0

(C) Odds > 1

Odds and probabilities

Definition: Let π denote the probability an event happens (e.g., $\pi = P(\text{Acceptance} = 1)$). Then the *odds* are

$$\frac{\pi}{1 - \pi} \quad \frac{0.5}{1 - 0.5} = 1 \quad \frac{0.6}{1 - 0.6} = \frac{0.6}{0.4} = 1.5 > 1$$

If the probability of an event is > 0.5 , what is true about the odds?

(A) Odds > 0.5

(B) Odds > 0

(C) Odds > 1

Answer: Odds > 1

Odds and probabilities

Definition: Let π denote the probability an event happens (e.g., $\pi = P(\textit{Acceptance} = 1)$). Then the *odds* are

$$\frac{\pi}{1 - \pi}$$

True or false: the odds increase when the probability increases.

Odds and probabilities

Definition: Let π denote the probability an event happens (e.g., $\pi = P(\text{Acceptance} = 1)$). Then the *odds* are

$$\frac{\pi}{1 - \pi}$$

True or false: the odds increase when the probability increases.

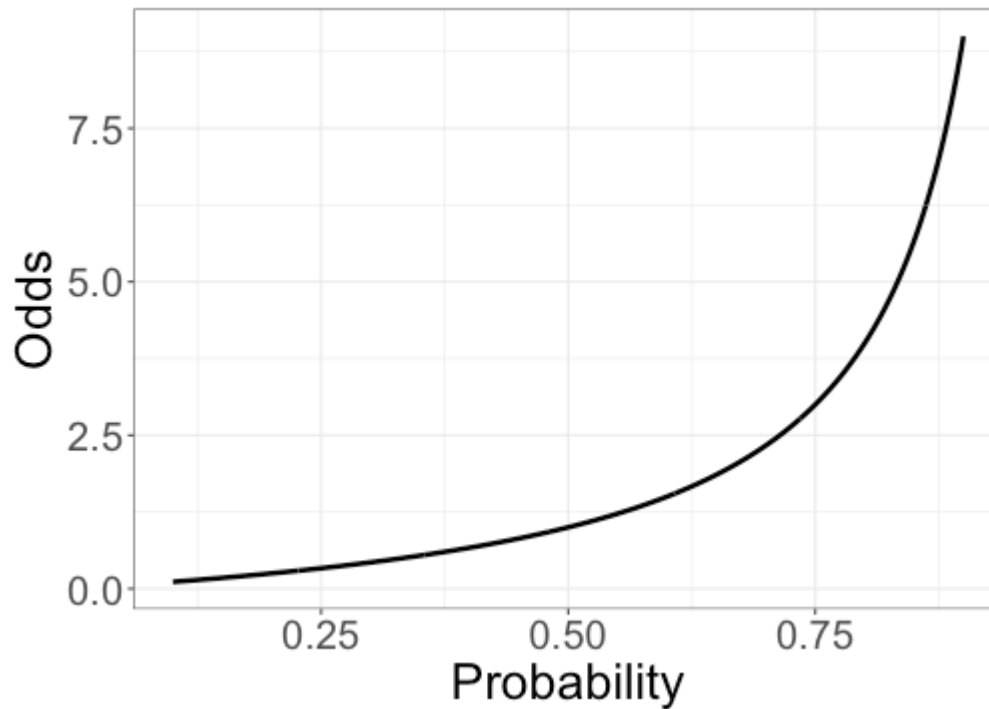
Answer: True

$$\text{as } \pi \uparrow, \quad \frac{\pi}{1 - \pi} \uparrow$$

odds is a monotonically increasing
function of probability

Odds as a function of probability

$$\text{Odds} = \frac{\pi}{1 - \pi}$$



Odds

$$\text{Odds} = \frac{\pi}{1 - \pi}$$

Probabilities are between 0 and 1. What range of values can odds take?

(A) between 0 and 1

(B) between 0 and ∞

(C) between $-\infty$ and ∞

Odds

$$\text{odds} \in [0, \infty)$$

$$\log: [0, \infty) \rightarrow (-\infty, \infty)$$

$$\log(x) \rightarrow -\infty \text{ as } x \rightarrow 0$$

$$\log(x) \rightarrow \infty \text{ as } x \rightarrow \infty$$

$$\text{Odds} = \frac{\pi}{1 - \pi}$$

Probabilities are between 0 and 1. What range of values can odds take?

(A) between 0 and 1

(B) between 0 and ∞

(C) between $-\infty$ and ∞

Answer: between 0 and ∞

$$\begin{array}{ll} \text{as } \pi \rightarrow 0, & \text{odds} \rightarrow 0 \\ \text{as } \pi \rightarrow 1, & \text{odds} \rightarrow \infty \end{array}$$

Log odds

Definition: Let π denote the probability an event happens (e.g., $\pi = P(\textit{Acceptance} = 1)$). Then the *log odds* are

$$\log\left(\frac{\pi}{1 - \pi}\right)$$

What range of values can log odds take?

(A) between 0 and 1

(B) between 0 and ∞

(C) between $-\infty$ and ∞

Log odds

Definition: Let π denote the probability an event happens (e.g., $\pi = P(\textit{Acceptance} = 1)$). Then the *log odds* are

$$\log\left(\frac{\pi}{1 - \pi}\right)$$

What range of values can log odds take?

(A) between 0 and 1

(B) between 0 and ∞

(C) between $-\infty$ and ∞

Answer: between $-\infty$ and ∞

Logistic regression

Logistic regression model: response is Acceptance

$$\pi = P(\text{Acceptance} = 1)$$

Our logistic regression model describes

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \text{GPA}$$

probability of acceptance to medical school, given GPA

+ log odds are between $-\infty$ and ∞ , so we can use a line!

Where did the ε term go?

π is not a random variable
(Acceptance is a random variable)

Logistic regression

Logistic regression model:

$$\pi = P(\textit{Acceptance} = 1)$$

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 \textit{GPA}$$

+ log odds are between $-\infty$ and ∞ , so we can use a line!

Where did the ε term go?

- + The ε term is used to include randomness in the model
- + The probability π *already* incorporates randomness
- + The probability and log odds are not random themselves; they describe randomness for the response

Fitted regression model $\hat{\pi}$, not π , b/c this is estimated

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

What are the estimated log odds of acceptance for a student with a 3.0 GPA?

Fitted regression model

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

What are the estimated log odds of acceptance for a student with a 3.0 GPA?

Answer: $-19.21 + 5.45 \cdot 3 = -2.86$

Fitted regression model

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

What are the estimated *odds* of acceptance for a student with a 3.0 GPA?

$$\begin{aligned} \underbrace{\frac{\hat{\pi}}{1 - \hat{\pi}}}_{\text{odds}} &= e^{-19.21 + 5.45 \text{ GPA}} \\ &= e^{-2.86} = 0.0573 \end{aligned}$$

Fitted regression model

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

$$\text{odds} = e^{\beta_0 + \beta_1 x}$$
$$\hat{\pi} = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

What are the estimated *odds* of acceptance for a student with a 3.0 GPA?

Answer: $\exp\{-2.86\} = 0.0573$

$$\frac{\hat{\pi}}{1 - \hat{\pi}} = 0.0573$$

$$\begin{aligned}\hat{\pi} &= (1 - \hat{\pi}) 0.0573 \\ &= 0.0573 - 0.0573 \hat{\pi}\end{aligned}$$

$$\Rightarrow \hat{\pi} + 0.0573 \hat{\pi} = 0.0573$$

$$\Rightarrow \hat{\pi} (1 + 0.0573) = 0.0573$$

$$\hat{\pi} = \frac{0.0573}{1 + 0.0573}$$

In general:

$$\text{odds} = \frac{\pi}{1 - \pi}$$
$$\Leftrightarrow \pi = \frac{\text{odds}}{1 + \text{odds}}$$